

Coding Challenge Guidelines

Please organize, design, test, document and deploy your code as if it were going into production, then send us a link to the hosted repository (e.g. Github, Bitbucket...).

Technical spec

The architecture will be split between a back-end and a web front-end, for instance providing a JSON in/out RESTful API. Feel free to use any other technologies provided that the general client/service architecture is respected. Choose one of the following technical tracks that best suits your skillset or the role that you are applying:

1. **Full-stack**: include both front-end and back-end.
2. **Back-end track**: include a minimal front-end (e.g. a static view or API docs). Write, document and test your API as if it will be used by other services.
3. **Front-end track**: include a minimal back-end, or use the data service directly. Focus on making the interface as polished as possible.

Back-end

We believe there is no one-size-fits-all technology. Good engineering is about using the right tool for the right job, and constantly learning about them. Therefore, feel free to mention in your README how much experience you have with the technical stack you choose, we will take note of that when reviewing your challenge. Here are some technologies we are more familiar with:

- Python, Go, Typescript

You are also free to use any web framework. If you choose to use a framework that results in boilerplate code in the repository, please detail in your README which code was written by you (as opposed to generated code).

Front-end

The front-end should ideally be a single page app. You may take this opportunity to demonstrate your UI knowledge.

- It can be developed either using Typescript or Flutter.

Functional spec

Prototype one of the following projects:

1. Departure Times
2. SF Movies
3. Food Trucks

The UX/UI is totally up to you. If you like, get creative and add additional features a user might find useful!

Departure Times

Create a service that gives real-time departure time for public transportation (use freely available public API). The app should geo localize the user.

The data is available on [Transport for London Unified API](#) (London)

SF Movies

Create a service that shows on a map where movies have been filmed in San Francisco. The user should be able to filter the view using autocomplete search.

The data is available on [DataSF: Film Locations](#).

Food Trucks

Create a service that tells the user what types of food trucks might be found near a specific location on a map.

The data is available on [DataSF: Food Trucks](#)

Host it!

When you're done, host it somewhere (e.g. on Amazon EC2, Heroku, Google AppEngine, etc.).

Our Evaluation Criteria We will assess the following aspects of your solution:

- Knowledge of the tech stack you choose.
- Proper structure of the application layers.
- Handling of outgoing calls.
- Effective use of environment variables.
- Implementation of unit tests.
- Logging mechanisms.
- Error handling strategies.
- Documentation quality.
- Code organization, module separation, readability, and comments.
- Commit history.
- Demo

Artifacts to be submitted are as below:

1. Github repository link of the source code
 - a. The github repo should contain the Readme.md file with all the details & documentation
 - b. The readme.md should also contain the details of the developer
2. screen capture videos that shows the recording of the app/webpage running and the functionalities used (on an emulator or browser or real device)

Please email us at To HR@kjbnlabs.in CC siddhika.patel@kjbnlabs.in